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SUPERSEDING
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MILITARY SPECIFICATION

RECEIVING SET, RADIO AN/URR-71

This specification is approved for use by the Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification defines the requirements for the following units of a general purpose very high frequency radio receiving set designated Receiving Set, Radio AN/URR-71:

Receiver, Radio R-1518/UR
Antenna, Whip AS-2887/UR
Field Pack, Canvas CW-1005/UR

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, US Army Communications-Electronics Command, ATTN: DRSEL-ED-TO, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5820

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1.2 Description. Radio Receiver R-1518/UR, is a superheterodyne receiver capable of receiving AM, FM, CW and MCW signals, in two bands over a frequency range of 19.0 to 157.5 MHz. The receiver is powered by 115/230V, 47 to 400 hertz ac, 26Vdc vehicular, or an internally contained dry battery power source, consisting of 12 type BA-30/U or BA-3030/U cells. The basic receiver package also includes a whip antenna AS-2887/UR and Canvas Field Pack CW-1005/UR.

2. APPLICABLE DOCUMENTS.

2.1 Issues of documents. The following documents form a part of this specification to the extent specified herein. The effective issue or revision of specifications and standards shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, unless (a) specific issues are set forth herein, or (b) different issues than those specified in this specification are set forth in the solicitation. The date of the applicable DODISS and supplements thereto, and the effective issue or revision of drawings cited herein, shall be as specified in the solicitation or contract.

SPECIFICATIONS

FEDERAL

CCC-C-419	Cloth, Duck, Cotton, Unbleached, Plied-Yarns, Army and Numbered

MILITARY

MIL-P-116	Preservation-Packaging, Methods of
MIL-L-3891	Luminescent Material and Equipment (Nonradioactive)
MIL-R-10273	Radio Sets, AN/PRC-8( ), AN/PRC-9( ), AN/PRC-10( )
MIL-P-11268K	Parts, Materials and Processes Used In Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-C-39012	Connector, Coaxial, Radio Frequency, General Specification for. Also include supplements /21 and /31
MIL-R-55317	Receiving Set, Radio AN/URR-70 and Kit, Accessory MK-1517/UR

STANDARDS

FEDERAL

FED-STD-751	Stitches, Seams and Stitchings

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SPECIFICATIONS (Cont'd)

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461A w/Notice 4	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462 w/Notice 3	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-781C	Reliability Tests, Exponential Distribution
MIL-STD-810C	Environmental Test Methods
MIL-STD-1275	Electrical Circuit, 28 Volt DC, Transient Characteristics for Military Vehicles
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities

DRAWINGS

U.S. Army Electronics Research and Development Command

DL-SC-B-91304	Antenna AT-272( )/PRC
SC-A-078020	Burn-In Procedure for Radio Receivers R-1218/UR and R-1518/UR
SM-A-747539	Test Procedure-Receiver, Radio R-1518/UR
DL-SM-B-747551	Antenna AS-2887/UR
DL-SM-B-747558	Field Pack, Canvas CW-1005/UR
DL-SM-B-747560	Receiving Set, Radio AN/URR-71
DL-SM-B-747564	Maintenance Item Set
SM-B-747787	Lamp, Incandescent, Modified

SPECIAL INSTRUCTIONS

SPI 1G00075	Special Packaging Instructions (see para. 5.1).

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies).

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3. REQUIREMENTS.

3.1 Basic receiver complement. The AN/URR-71 receiving set consists of the following listed items only, and requires additional items to supply audio response and power (see 3.2 below).

a. Receiver, Radio R-1518/UR
b. Antenna, Whip AS-2887/UR
c. Field Pack, Canvas CW-1005/UR
d. Running spare parts in pocket of field pack cover, including:

(1) Fuses, cartridge, 4 ea., Part No. F02A250V1-8A
(2) Lamp, Incandescent, Modified, 4 ea., Part No. SM-B-747787

3.2 Ancillary items. The following ancillary items are not part of the basic receiving set, but are used with it.

3.2.1 Kit, Accessory MK-1517/UR. The accessory kit is specified in MIL-R-55317(ER) and contains a shock mount for use in vehicular operation, power cables and an extender cable for test and diagnostic work.

3.2.2 Maintenance item set. The maintenance item set contains those special purpose extender cables, tuning tools, connectors, etc., required for DS/GS level maintenance of the R-1518/UR receiver, which capability is not available as standard test equipment. The maintenance item set is specified in MIL-R-55317(ER).

3.2.3 Audio output device. The Receiving Set AN/URR-71 does not include earphones or a speaker as an audio output transducer. The audio output of the receiving set is a 600 ohm termination.

3.2.4 Internal batteries. As well as being powered by 115/230 Vac and 26 Vdc, the R-1518/UR receiver may be powered by internal dry cells. The battery source is 18 Vdc nominal, supplied by 12 each BA-30/U or BA-3030/U dry cells.

3.3 Burn-in. Each receiver to be delivered on contract, or to be subjected to first article tests or reliability tests in accordance with this specification, shall be subjected to a 48 hour burn-in to limit infant mortality. The last 24 hour burn-in period shall be failure free. This burn-in shall be prior to any or all tests specified herein. (see 4.3).

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3.4 First article. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.4 (see 6.2.d and 6.2.f).

3.5 Mechanical and construction requirements. The equipment shall be constructed in accordance with the requirements of this specification and drawing DL-SM-B-747560 which describes the AN/URR-71 Receiving Set, including the requirements for parts, materials and processes thereon.

3.5.1 Parts, materials and processes; general. In addition to the requirements of this specification, the general requirements of MIL-P-11268K(Army) including the selection requirements therein, shall apply.

3.5.1.1 Finish. The equipment shall be finished in accordance with MIL-F-14072(EL) and the equipment drawings. The final paint film on all exterior surfaces shall be green color (olive drab) semigloss enamel matching a color chip provided by the procuring agency (see 6.4).

3.5.1.2 Marking.

3.5.1.2.1 General. Marking shall conform to MIL-M-13231(EL) (see 4.5). Front panel marking shall be Group I as described in MIL-M-13231(EL). Location of marking shall be as specified on the equipment drawings.

3.5.1.2.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.1.2.3 Luminescent marking. Luminescent marking shall be type P in accordance with Specification MIL-L-3891.

3.5.1.2.4 Serial numbers. Each Radio Receiver R-1518/UR shall be serially numbered. The format of the serial number and the starting number will be specified in the procurement document.

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3.5.1.3 Fabric and thread.

3.5.1.3.1 General. The color of fabric shall be Olive Drab No. 7. This requirement shall be met immediately after treatment. The color of thread shall match shade S-1 of the United States Army Standard Color Card for Official Standardization Shades of Sewing Thread (see 6.5). Fabric and thread shall be preshrunk, or allowance shall be made for shrinkage in order to provide for satisfactory fit of finished items both before and after they are soaked in tap water at normal room temperature for 24 hours and then dried thoroughly at normal room temperature. Specification and treatment of materials shall be in accordance with paragraphs 3.8.3 and 3.9.7 of MIL-P-11268K(Army).

3.5.1.3.2 Duck. Cotton duck shall be in accordance with Federal Specification CCC-C-419, Type 1, Number 6 duck.

3.5.1.3.3 Stitches, seams, and stitching. Stitches, seams, and stitching shall conform to Federal Standard FED-STD-751.

3.5.1.4 Tropicalization (fungus-inert materials). No fungus nutrient materials, treated or otherwise, shall be used except in a hermetically sealed enclosure for the Receiver R-1518/UR. Fungus-inert materials shall be in accordance with Requirement 4 of MIL-STD-454 (see 3.7.9 and 4.9.14).

3.5.2 Size and weight. The size of the receiver shall be 5 $\pm$ 1/16 inches high by 7 1/8 $\pm$ 1/16 inches wide x 9 3/8 $\pm$ 1/16 inches long. The weight shall not exceed 11 pounds (without batteries).

3.5.3 Tuning control. Two speed tuning shall scan each entire bandwidth in approximately 12 revolutions at the fast tuning rate. The slow tuning rate shall be one-fourtieth of the fast tuning rate $\pm$ 20%. The mechanical stop at each end of the tuning range shall be capable of withstanding 10 inch-pounds of torque.

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3.5.4 Whip Antenna, AS-2887/UR. The whip antenna shall be constructed to and meet the requirements of DL-SM-B-747551. It is made from an Antenna AT-272( )/ PRC, conforming to Specification MIL-R-10273 and drawing number DL-SC-B-91304.

3.5.5 Field Pack, Canvas, CW-1005/UR. A water repellant, cotton duck, zipperclosing, carrying case meeting the requirements of 3.5.1.3 and DL-SM-B-747558 shall be supplied to house each receiver.

3.5.6 Equipment safety engineering.

3.5.6.1 Personnel hazards. Personnel hazards shall be kept to a minimum by control of configuration changes and parts selection to assure compliance with Requirement 1 of MIL-STD-454. Compliance with these requirements shall be verified through a visual inspection (see 4.7).

3.5.6.2 Edge rounding, exposed. Exposed edges shall be rounded to a minumum radius of 0.04 inch (1 mm) and exposed corners to a minumum of 0.03 inch (0.75 mm).

3.5.7 Human engineering. Configuration design criteria affecting human factors operation, maintainability and handling shall be maintained through compliance with MIL-STD-1472.

3.5.8 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable without modification of such items or the system and shall conform to Requirement 7 of MIL-STD-454 (see 4.10). Individual items shall not be selected for fit or performance, however, matched pairs or sets, when permitted, may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating or characteristic.

3.6 Electrical requirements.

3.6.1 Antenna inputs.

3.6.1.1 Input impedance. The ANT 2 antenna input terminal shall match a nominal 50 ohm impedance and the ANT1 terminal shall match a high impedance affixable whip. The antenna selector switch shall select the antenna input for use and shall ground the unused connector.

3.6.1.2 Antenna input isolation. The isolation between the antenna inputs shall be greater than 70 db.

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3.6.2 Frequency range (overlap). The frequency range of the receiver shall be 19.0 to 157.5 MHz, covered in 2 bands as follows:

Band	Frequency range (MHz)
1	19.0 to 52.5
2	47.5 to 157.5

3.6.3 Frequency stability.

3.6.3.1 Frequency stability vs. time. When operating from a cold start, the frequency stability of the local oscillator shall not deviate more than 0.015% in the first 5 minutes nor more than 0.025% from 5 to 60 minutes after the initial start-up.

3.6.3.2 Frequency stability vs. internal battery voltage. As the battery supply voltage is varied from 18V to 14V the receiver frequency for any given dial setting shall not deviate more than ± 0.02%.

3.6.3.3 Frequency stability vs. received signal level. The observed frequency of maximum response, when receiving a signal of 0.5V at the antenna input, shall not deviate more than ± 5 kHz from that observed at threshhold sensitivity, 1 microvolt input.

3.6.3.4 Frequency stability vs. temperature. When tested in accordance with 4.8.2/4.9.1 the frequency drift of the receiver shall not exceed 0.5% over the operating temperature range. The rate of frequency drift shall not exceed 0.006% per degree F change in temperature.

3.6.3.5 Frequency stability vs. humidity. The frequency drift of the receiver shall not exceed 0.125% throughout the humidity cycle when tested in accordance with 4.8.2/4.9.2.

3.6.4 Sensitivity.

3.6.4.1 FM, AM and CW sensitivity. The signal level required to yield the signal plus noise to noise ratio specified below, shall not exceed the input tabulated below, when measured with a 10 mW output reference level.

	Signal Description	IF Bandwidth	S + N / N	Input Signal Level
CW Sensitivity:	Carrier Freq. per 4.8.2 Modulation none.	10 kHz	10 db	0.4 µV

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3.6.4.1 FM, AM and CW sensitivity (Cont'd).

	Signal Description	IF Bandwidth	S + N / N	Input Signal Level
AM Sensitivity	Carrier Freq. per 4.8.2 Modulation 1000 Hz, 30%, AM	10 kHz	10 db	1.2 μV
FM Sensitivity	Carrier Freq. per 4.8.2 Modulation 1000 Hz FM, 15 kHz deviation	75 kHz	20 db	1.2 μV

3.6.4.2 AM sensitivity (Whip antenna input). The receiver shall have a 10db or better signal-plus-noise-to-noise ratio, with an output reference level of 10 mW when the following 1000 Hz, 30% modulated, AM signal levels are impressed on the whip antenna input under the test conditions of 4.8.2.

Band	Frequency (MHz)	Signal Level, μV
1	24	8.0
1	35	6.0
1	46	5.0
2	60	4.0
2	100	3.0
2	140	3.0

3.6.4.3 AM sensitivity vs. internal battery voltage. AM sensitivity of the receiver, when tested under the conditions of 3.6.4.1 above, shall not require more than a 2.0 μV signal input level to maintain a 10db signal-plus-noise-to-noise ratio, with a 10 mW reference output, when the battery supply voltage is reduced from 18V to 14V.

3.6.4.4 AM and CW sensitivity vs. temperature. At the operating temperature extremes, -40° and +154°F, the signal level required to produce a signal-plus-noise-to-noise ratio of 10 db, with an output reference level of 10mW shall not exceed 2.0μV for AM operation and 0.7μV for CW operation.

3.6.4.5 FM sensitivity vs. temperature. At the operating temperature extremes, -40° and +154°F, the signal level required to produce a signal-plus-noise-to-noise ratio of 20 db, with an output reference level of 10 mW shall not exceed 2.0 μV for FM operation.

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3.6.4.6 AM and CW sensitivity vs. humidity. Throughout the humidity cycle, the signal level required to produce a signal-plus-noise-to-noise ratio of 10db, with an output reference level of 10 mW shall not exceed 2.0 μV for AM operation and 0.7 μV for CW operation.

3.6.4.7 FM sensitivity vs. humidity. Throughout the humidity cycle, the signal level required to produce a signal-plus-noise-to-noise ratio of 20 db, with an output reference level of 10 mW shall not exceed 2.0 μV for FM operation.

3.6.5 Calibration oscillator.

3.6.5.1 Frequency calibrator accuracy. The crystal controlled frequency calibrator shall have a fundamental frequency of 2 MHz ± 0.005%. The calibration oscillator shall be activated when the function switch is placed in the calibrate position.

3.6.5.2 Frequency calibrator degradation. The calibration oscillator frequency shall not vary more than ± 0.0075% under the various environmental conditions when tested in accordance with 4.82.

3.6.5.3 Injection level. The minimum permissible calibration oscillator injection shall be sufficient to yield a half scale deflection on the panel meter when the RF gain control is at maximum, while the maximum permissible injection shall not cause more than a half scale meter deflection when the RF gain control is set at minimum.

3.6.5.4 Spurious indications. There shall be no observable panel meter readings as the result of spurious calibration oscillator emissions when observed at maximum RF gain.

3.6.5.5 Calibration adjustment. The calibration cursor shall be capable of being adjusted by means of the CAL ADJ Control both above and below the Calibration Frequency Mark on the dial at each calibration point in all five bands.

3.6.6 Dial readout error. Calibration adjustment at any check point shall result in no more than 1 minor division (Band 1) and 0.15% or 2 minor divisions (Band 2), whichever is greater, readout error at any point between that check point and adjacent calibration points. One check point in each band is allowed to exceed the above specified limit.

3.6.7 Tuning backlash. The receiver tuning shall be directly calibrated in frequency. The control shall operate free from backlash and have a resetability of 0.1 percent.

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3.6.8 Spurious Responses.

3.6.8.1 Image rejection. Image rejection shall be suppressed at least 80 db from 19 MHz to 52.5 MHz in Band 1 and at least 80 db at 47.5 MHz, 60 db at 102 MHz, and 50 db at 156 MHz in Band 2.

3.6.8.2 IF rejection. IF rejection shall be suppressed at least 80 db.

3.6.8.3 Crossmodulation. Crossmodulation response shall be suppressed by the following amounts:

Generator 1 Reference Signal	Separation between desired and undesired signals	Generator 2 Signal Level
124 MHz, 1.2 V	± 0.5 MHz	.12 mV (40 db down)
124 MHz, 1.2 V	± 1.25 MHz	1.2 mV (60 db down)
124 MHz, 1.2 V	± 3.75 MHz	12.0 mV (80 db down)

3.6.8.4 Intermodulation distortion products. The intermodulation distortion products shall be suppressed at least 66 db except that in-band third order distortion products shall be suppressed at least 60 db.

3.6.9 Intermediate frequency amplifier.

3.6.9.1 IF center frequency. The IF center frequency shall be 10.7 MHz ± 0.1%.

3.6.9.2 IF selectivity. The receiver IF selectivity shall be as specified below. Response characteristics shall be free of irregularities in excess of 3 db within 60 db of maximum response.

IF Response Characteristics

Response (db down from max)	Narrow Bandwidth 10 kHz Position (kHz)	Broad Bandwidth 75 kHz Position (kHz)
reference	- - - - - - - - -	- - - - - - - - -
6	9 to 11	70 to 80
60	Less than 30	Less than 225
80	Less than 60	Less than 450

3.6.10 AGC characteristic. The AGC action of the receiver shall limit output variations as follows:

Input Signal	Output Ref. Level	Output Change Limit
2.5µV to 0.1V	1 mW	6 db
0.1V to 0.5V	1 mW	10 db

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3.6.11 Limiter characteristics. The audio output shall not vary more than 6 db when signal levels from 2.5μV to 0.5V are applied to the antenna input and the receiver is operated in the FM position with AGC on.

3.6.12 Beat frequency oscillator. The frequency of the BFO shall be 10.7 MHz ± 0.05%. The BFO shall operate when the function switch is in both the CW and the calibrate position.

3.6.13 Audio output.

3.6.13.1 Audio power output. The audio power output shall not be less than 20 mW with 18Vdc applied to the internal battery supply terminals. When the supply voltage is reduced to 14Vdc, the output shall not be less than 10mW.

3.6.13.2 Audio harmonic distortion. With a 10uV AM signal and output adjusted to 10 mW audio harmonic distortion plus noise shall not exceed 10%.

3.6.13.3 Audio response. With 1000 cps as a reference level the overall audio response shall be as follows:

Audio frequency (Hertz)	Response (db)
300	-3 ± 2
1000	0
2500	-3 ± 2
7000	-16 or greater

3.6.13.4 Microphonics. There shall be no objectionable microphonic noises greater than 3 db above the receiver quiescent value. An objectionable microphonic noise is defined as an audible interference, sustained or repetitious, which can affect the intelligibility of an incoming signal for a period longer than 3 seconds.

3.6.14 Diode output.

3.6.14.1 Diode output level. With a 1.2μV AM signal 30% modulated at 1000 Hz applied to the 50 ohm antenna input, the diode output shall not be less than 2.0V as measured across a 50,000 ohm resistive load. The diode output shall be negative with respect to ground. The input signal level shall be 2.0 μV when testing for temperature and humidity requirements.

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3.6.15 Discriminator output. The FM discriminator shall have adequate bandwidth to produce distortion free output when operated in the 75 kHz IF bandwidth. Output of the discriminator shall feed the audio amplifier as well as a front panel receptacle labeled DISCR. The discriminator output shall provide a level of at least 50 mV across 600 ohms when the receiver is fed with a 1.2 uV, 1000 Hz modulated FM signal having 15 kHz deviation. The bandpass of the discriminator shall be 250 Hz to 16,000 Hz, with no inflection of more than 2 db. Connection of a load to the discriminator output shall not disturb the audio output of the receiver.

3.6.16 Power supplies. The receiver shall be capable of operating from internally contained dry batteries and an internal power converter operated from a 26Vdc vehicular source or a 115/230V, 47 to 400 Hz source, selected by a switch located on the rear of the receiver.

3.6.16.1 Input power requirement. The total power drain shall not exceed 1.5 W for internal battery operation, 3.0 W for vehicular operation and 5.5 W for 115/230 Vac operation. The input power shall be measured with dial light off.

3.6.16.2.1 Voltage regulation. The output voltage of the power converter shall be regulated to 11.2 ± 0.2 Vdc over the full range of AC,battery and vehicular inputs as follows:

Input Frequency	Input Voltage
60 Hz	200 to 250 V and 90 to 130 V
400 Hz	200 to 250 V and 90 to 130 V
dc (vehicular)	22 to 32 V
dc (battery)	14 to 20 V

3.6.16.2.2 Output voltage ripple. The output voltage ripple of the receiver power supply shall not exceed 10 mV rms when measured at the diode output.

3.6.16.3 Battery supply. The receiver shall use twelve 1½ V standard dry cells, type BA-30/U or BA-3030/U, as an internal battery supply. Batteries shall not be supplied with the equipment.

3.6.16.4 Solid state device protection. All solid state circuitry shall be protected from destructive transients originating in the vehicular supply under the worst case conditions described in MIL-STD-1275, or originating in any other supply. This protection shall be accomplished by the use of semiconductor devices specifically designed for transient suppression in this particular application.

3.6.16.5 Reverse polarity protection. The receiver shall be protected against accidental reverse polarity. It shall not be damaged by reverse polarity input.

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3.6.16.6 Panel meter. A meter on the front panel of the receiver shall monitor the dry battery voltage level, the internal power converter output voltage level, and the carrier level. The meter shall have negligible effect on the metered circuits. Full scale deflection during AGC operation shall indicate approximately a 0.5V input signal. The scale markings shall be logarithmic. No more than 10 percent deflection shall be indicated with no signal input. During manual gain control operation, full scale deflection shall indicate the beginning of an overload condition for AM operation or a signal 40 db above limiting for FM operation. The meter shall be marked with "Good" and "Replace", color coded red and green, and shall be calibrated such that the upper limit of the green area is 18V and the lower limit of this area is 14V.

3.6.17 Dial light. With all external power input options, the dial light shall operate when the dial light switch is in the ON or the M-ON position. Selection of the internal battery power input option shall limit dial light illumination to dial light switch position M-ON.

3.6.18 Final operation inspection. Immediately prior to preparation for delivery, each receiver shall be given the "Final Operation" inspection as specified in 4.8.2.

3.7 Environmental requirements. The equipment shall meet the following service conditions. Where a test is referenced, meeting the performance criteria shall be considered as compliance with the requirement.

3.7.1 Temperature.

3.7.1.1 Low temperature. The receiver shall be capable of operation without degradation at ambient temperatures as low as -40°F (-40°C) and of being stored at temperatures as low as -65°F (-54°C). Before, during and after testing as specified in 4.9.1.1, the receiver shall meet the following requirements:

Paragraph	Measurement
3.6.3.4	Freq Stability vs. Temp.
3.6.4.4	AM & CW Sensitivity vs. Temp.
3.6.4.5	FM Sensitivity vs. Temp.
3.6.5.2	Freq Calibrator Degradation
3.6.5.3	Injection Level
3.6.10	AGC Characteristic
3.6.11	Limiter Characteristic
3.6.14	Diode Output

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3.7.1.2 High temperature. The receiver shall be capable of operation without degradation at ambient temperatures as high as +154°F (68°C) and of being stored at temperatures as high as +160°F (71°C). Before, during and after testing as specified in 4.9.1.2 the receiver shall meet the same requirements as specified for low temperature.

3.7.2 Humidity. The receiver shall sustain no physical damage and shall be capable of operation at high temperature (86°F) and in high humidity (94%). Before, during and after testing as specified in 4.9.2 the receiver shall meet the performance specified in the tabulation below. After the humidity exposure, the receiver shall exhibit no physical damage such as corrosion, rust, blistering, swelling or deterioration of parts and materials.

Paragraph	Measurement
3.6.3.5	Freq Stab vs. Humidity
3.6.4.6	AM & CW Sensitivity vs. Humidity
3.6.4.7	FM Sensitivity vs. Humidity
3.6.5.2	Freq Cal Degradation
3.6.5.3	Injection Level
3.6.10	AGC Characteristic
3.6.11	Limiter Characteristic
3.6.14	Diode Output

Injection Level (3.6.5.3) Test will be conducted as a Prehumidity and Post Humidity Requirement only.

3.7.3 Altitude. The receiver shall be capable of operation at 15,000 feet altitude and of being air transported at 40,000 feet. Before, during and after testing as specified in 4.9.3, the receiver shall meet full specified performance for the following measurements:

Paragraph	Measurement
3.6.5.1	Frequency Calibrator Accuracy
3.6.5.3	Injection Level

In addition, the receiver shall meet full specification performance for the following measurements before and after the testing specified in 4.9.3:

Paragraph	Measurement
3.6.4.1	FM, AM & CW Sensitivity

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3.7.4 Vibration and shock. The equipment shall be able to withstand the dynamic shock and vibration stresses normally encountered in servicing, in transportation by rail, air, sea, truck or semitrailer, or as loose truck cargo, or in vehicular use, without degradation in performance or serious physical damage. The equipment shall meet the requirements of TABLE I before and after testing as specified in 4.9.4 through 4.9.9.

TABLE I - Vibration and shock requirements.

Test	Test Para.	Performance	Notes
Vibration (Ground vehicle installation)	4.9.4	Specified performance No physical damage	A, E
Shock (basic design)	4.9.5	Specified performance No physical damage	A, E
Vibration (Secured cargo transportation)	4.9.6	Specified performance No physical damage	C, E
Shock (Bench handling)	4.9.7	Specified performance No physical damage	D, F
Shock (loose cargo bounce)	4.9.8	Specified performance No physical damage, except for minor surface abrasions	B, E
Shock (transit drop)	4.9.9	Specified performance Minor physical damage only, incl. inoperative panel meter	B, E

(See TABLE I for Notes)

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TABLE I - Vibration and shock requirements (Cont'd)

NOTES:

A. Receiver shall be securely mounted on its shock mount, MT-4034/UR (p/o Accessory Kit MK-1517/UR). Test applies to receiver/mount assembly.

B. Receiver tested in its canvas bag. The bag may be severely damaged in test.

C. Receiver under test, alone, is secured to vibration table.

D. Unit under test is receiver alone.

E. Specified performance for these tests is interpreted as meeting the following criteria:

Para. 3.6.4.1 - AM, FM & CW Sensitivity at only one frequency in each band.

3.6.5.3 - Injection Level at only one frequency in each band.

3.6.14 - Diode Output at only one frequency in each band.

F. Specified performance for this test is interpreted as meeting the CW sensitivity requirement of 3.6.4.1 at one frequency only.

3.7.5 Rain. The receiver shall be capable of operation and shall show no evidence of physical damage or water leakage after exposure to rain accompanied with high wind as an anticipated service condition. Before and after testing as specified in 4.9.10, the equipment shall meet full specification performance for the following measurements:

Paragraph	Measurement
3.6.4.1	AM and FM Sensitivity
3.6.5.3	Injection Level

3.7.6 Immersion. Before and after testing as specified in 4.9.11, the receiver shall show no evidence of water leakage and shall meet full specified performance for the following measurements:

Paragraph	Measurement
3.6.4.1	AM and FM Sensitivity
3.6.5.3	Injection Level

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3.7.7 Dust. The receiver shall be capable of operation and shall show no physical damage after exposure to fine blowing dust. Before and after testing as specified in 4.9.12, the equipment shall meet full specification performance for the following measurements:

Paragraph	Measurement
3.6.4.1	AM and FM Sensitivity
3.6.5.3	Injection Level

3.7.8 Salt fog. The equipment (including all accessories) shall be capable of operation after exposure to a marine or salt laden atmosphere. After testing as specified in 4.9.13, the equipment shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface. No rust shall be imposed on the test item by the test facility. The equipment shall meet full specification performance for the following measurements:

Paragraph	Measurement
3.6.4.1	AM and FM Sensitivity
3.6.5.3	Injection Level

3.7.9 Fungus. The equipment (including accessories) shall show no evidence of viable fungus or corrosion when subjected to the test specified in 4.9.14. Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface. Operation of the equipment is not required.

3.8 Electromagnetic interference characteristics. The receiver shall meet the electromagnetic interference requirements specified in MIL-STD-461A, Notice 4, Table A-I-MIL-E-55301, for class CE equipments. Tests and test methods shall be as specified in MIL-STD-462(Notice 3(EL)), except that tuned channel frequencies shall be excluded for power line conducted susceptibility measurements of CS02. If the contractor considers it desirable to modify the receiver in any manner to facilitate conformance to the specified test limits, he shall obtain the approval of the contracting officer before proceeding with the modification.

3.9 Reliability. The reliability of the equipment shall be such that it is capable of meeting a lower test MTBF of 720 hours and an upper test MTBF of 2160 hours as defined in MIL-STD-781. During testing as specified in 4.11 the equipment shall meet the performance specified below:

a. Noise output of the receiver shall be constantly monitored during the length of the test. Cessation of noise output indicates failure.

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3.9 Reliability (Cont'd).

b. In addition to the above, the receiver shall meet full specified performance for the following measurements:

Paragraph	Measurement	Measurement Frequency
3.6.4.1	AM Sensitivity	Every 3rd day } alternating
3.6.4.1	CW Sensitivity	Every 3rd day } days
3.6.4.1	FM Sensitivity	Every 3rd day }
3.6.5.3	Injection Level	Daily
3.6.3.1	Freq Stability vs. Time	Every week
3.6.5.1	Freq Calibrator Accuracy	Every week
4.6.1	All Group A tests	Every 3rd week

3.10 Workmanship. The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

In MIL-P-11268K (Army)

General requirements for plastic material and plastic parts
Wiring and cabling, including:
  Slack
  Protection
  Clearance
  Splicing
Connections, general
Grounding
  Wire and cable shielding
Flux and cleaning agent for soldering
Process for soldering
Cleaning
Riveting
General requirements for securing of parts
Cleaning after welding
Mechanical operation of controls
Process for welding
Self-tapping screws

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (does not include packaging (see 4.4)).

b. Inspection covered by subsidiary documents (see 4.5).

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before packaging (see 4.6).

(2) Quality conformance inspection of packaging (see 4.12).

4.3 Burn-in. Prior to the start of any or all of the following tests, each receiver shall be burned-in in accordance with SC-A-678020 and shall meet the requirements of that procedure and 3.3 herein.

4.4 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.4.1 First article units. Unless otherwise specified in the contract or purchase order, the contractor shall furnish four first article units of the complete equipment on order.

4.4.2 First article inspection. The first article inspection shall consist of inspections specified in TABLE II and shall be performed in the order specified herein.

4.4.3 First article data. The first article test plan and test report shall be as required in the contract or purchase order (see 6.2.f).

4.5 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection requirements specified in the contract or purchase order.

Item	Where Required
Parts, materials, and processes; general	3.5.1
Finish	3.5.1.1
Marking	3.5.1.2
Safety	3.5.6
Human Engineering	3.5.7

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TABLE II - First article inspection

Inspections	Req. Para.	Test Para.	Number of Receiver Samples Three Units UNIT 1	UNIT 2	UNIT 3	Four Units UNIT 1	UNIT 2	UNIT 3	UNIT 4
1. Inspection covered by subsidiary documents.			Inspections to be performed on all units.						
2. Group A inspection	See TABLE III		Inspections to be performed on all units.						
3. Group B inspection	See TABLE IV		Inspections to be performed on all units.						
4. Group C inspection	As follows:								
Electrical/Mechanical									
Solid state device protection	3.6.16.4	4.8.2	1			1			
Reverse polarity protection	3.6.16.5	4.8.2	2			2			
Canvas shrinkage	3.5.1.3	4.9.15	3			3			
Tuning control	3.5.3	4.9.16	4			4			
EMI	3.8	4.9.17	5			5			
Environmental									
Temperature	3.7.1	4.9.1		1			1		
Humidity	3.7.2	4.9.2	6					1	
Altitude	3.7.3	4.9.3		2			2		
Vibration (ground vehicle)	3.7.4	4.9.4		5			4		
Shock (basic design)	3.7.4	4.9.5		6			5		
Vibration (secured cargo)	3.7.4	4.9.6			3				3
Bench handling	3.7.4	4.9.7			4				4
Loose cargo bounce	3.7.4	4.9.8		7				4	
Transit drop	3.7.4	4.9.9			5				5
Rain	3.7.5	4.9.10		3			3		
Immersion	3.7.6	4.9.11			1			2	
Dust (Note C)	3.7.7	4.9.12	7						1
Salt fog (Note C)	3.7.8	4.9.13		4					2
Fungus (Note C)	3.7.9	4.9.14			2			3	
5. Group D inspection Reliability	Receivers selected as required to meet 3.9 and 4.11								

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TABLE II - First article inspection (Cont'd)

NOTES:
A. The inspections 1 through 3, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements specified in the table.

B. The numbers in the unit columns in the table covered by inspection 4 specify the order of inspections for the indicated unit. Subject to Government approval, the order of non-damaging test and the choice of which units are subjected to the non-damaging tests may be varied at the contractors convenience.

C. The equipment shall be thoroughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent tests.

4.6 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.5 and 4.6.1 through 4.6.4 in that order. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.6.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in TABLE III. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subjected to sampling inspection, utilizing the procedures of MIL-STD-105, using general inspection level II and the AQL's indicated in TABLE III.

4.6.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government, except that the "Final Operation"inspection (see 3.6.18 and 4.8.2) shall be last.

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TABLE III - Group A inspection

Inspection	Req. Para.	Test Para.	AQL Major	AQL Minor
Visual and mechanical, General:	3.5, 3.10	4.5, 4.7		
Final assembly			1.0 dphu	4.0 dphu
Subassembly			1.0 dphu	4.0 dphu
Printed circuit boards			1.0%	4.0%
Modules			1.0%	4.0%
Electrical:				
Frequency range (overlap)	3.6.2	4.8.2		
AM sensitivity	3.6.4.1	4.8.2		
FM sensitivity	3.6.4.1	4.8.2		
CW sensitivity	3.6.4.1	4.8.2		
Freq. Calibrator accuracy	3.6.5.1	4.8.2		
Injection level	3.6.5.3	4.8.2		
Spurious indications	3.6.5.4	4.8.2		
Dial readout error	3.6.6	4.8.2		
Tuning backlash	3.6.7	4.8.2		
AGC characteristic	3.6.10	4.8.2	1.5%	*
Limiter characteristic	3.6.11	4.8.2		
Beat Freq. oscillator	3.6.12	4.8.2	for	
Audio power output	3.6.13.1	4.8.2		
Audio harmonic distortion	3.6.13.2	4.8.2	the	
Diode output level	3.6.14.1	4.8.2		
Discriminator output	3.6.15	4.8.2	group	
Input power requirements	3.6.16.1	4.8.2		
Voltage regulation	3.6.16.2.1	4.8.2		
Output voltage ripple	3.6.16.2.2	4.8.2		
Panel meter	3.6.16.6	4.8.2		
Dial light	3.6.17	4.8.2		
Final operation inspection	3.6.18	4.8.2		

*All electrical defects are considered major.

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TABLE IV - Group B inspection

Inspection	Req. Para.	Insp. Para.	AQL
Interchangeability - Electrical	3.5.8	4.10.1	1.0%
- Mechanical	3.5.8	4.10.2	4.0%
Input impedance	3.6.1.1	4.8.2 4.8.2.1	
Antenna input isolation	3.6.1.2	4.8.2	
Freq. stability vs. time	3.6.3.1	4.8.2	
Freq. stability vs. int. battery voltage	3.6.3.2	4.8.2	
Freq. stability vs. rec'd signal level	3.6.3.3	4.8.2	
AM sensitivity (whip antenna)	3.6.4.2	4.8.2	
AM sensitivity vs. int. battery voltage	3.6.4.3	4.8.2	
Calibration adjustment	3.6.5.5	4.8.2	4.0 dphu for the group
Image rejection	3.6.8.1	4.8.2	
IF rejection	3.6.8.2	4.8.2	
Crossmodulation	3.6.8.3	4.8.2	
Intermodulation distortion products	3.6.8.4	4.8.2	
IF center frequency	3.6.9.1	4.8.2	
IF selectivity	3.6.9.2	4.8.2	
Audio response	3.6.13.3	4.8.2	
Microphonics	3.6.13.4	4.8.2	
Battery supply	3.6.16.3	4.8.2 4.8.2.2	

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4.6.2 Group B inspection. Group B inspection shall be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. For the Group B interchangeability requirement, test samples may be separate units whose modules and replaceable parts were randomly selected and assembled prior to Group A testing. This inspection shall conform to TABLE IV and to the special procedures for small - sample inspection of MIL-STD-105.

4.6.2.1 Group B sampling plans. The Group B sampling plans for the AQL's listed in TABLE IV shall be S-4. All defects shall be considered major.

4.6.2.2 Order of inspection within Group B. Group B inspection shall be performed in an order satisfactory to the Government.

4.6.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in TABLE V. Samples shall be selected in accordance with 4.6.3.1.

4.6.3.1 Sampling for Group C inspection. Four samples of the complete equipment shall be selected at random, 1 sample for each Subgroup 1 through and including Subgroup 4 in TABLE V. The first samples shall be selected at the start of production from the first quality conformance inspection lot. These samples shall constitute the Group C requirement for the first 25 units produced. Thereafter, 4 samples of the complete equipment shall be selected at random, 1 sample for each Subgroup 1 through Subgroup 4 in TABLE V from every 75 units and fraction thereof produced. Salt fog and fungus shall not be repeated after the first production lot Group C inspection.

4.6.3.2 Order of inspection within Group C. Group C inspection shall be performed in any order acceptable to the Government; see NOTES A and C of TABLE II.

4.6.3.3 Group C failure. Action required relative to Group C failures shall be as specified in the contract or purchase order.

4.6.3.4 Reinspection of conforming Group C sample units. Unless otherwise specified sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A and Group B inspection.

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4.6.4 Group D inspection. This inspection shall consist of the tests specified in TABLE VI and shall be performed on units from lots which have been subjected to and met Group A and Group B inspection. Test conditions, methods and procedures shall be in accordance with paragraph 4.11. Individual units which have been subject to potentially damaging environmental tests will not be selected for reliability testing. Corrective action as a result of a failed Group C inspection may require that this inspection be reconducted.

4.6.4.1 Group D inspection lot formation. The inspection lot for production reliability acceptance testing shall be 100 units or fraction thereof produced, except that a final quantity of 42 or less may be added to a last lot of 100 if the procurement quantity is not an even multiple of 100.

4.6.4.2 Sampling for Group D inspection. The test sample for production reliability acceptance shall be seven (7) to ten (10) percent of the inspection lot, but not less than 3 units. It shall be randomly selected from the lot.

4.6.4.3 Group D failures. Actions required relative to Group D failures shall be as specified in the contract or purchase order.

4.6.4.4 Reinspection of conforming Group D sample units. Unless otherwise specified, sample units which have been subjected to and passed Group D inspection may be accepted on the contract or purchase order provided all visible and known damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.7 Visual and mechanical inspection. The equipment shall be visually and mechanically inspected to insure conformance to the construction requirements of 3.5 and the workmanship requirements of 3.10. Defects shall be classified in accordance with MIL-STD-252(EL).

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TABLE V - Group C inspection

Inspection	Req. Par.	Insp. Para.
Subgroup 1		
Solid state device protection	3.6.16.4	4.8.2
Reverse polarity protection	3.6.16.5	4.8.2
Canvas shrinkage	3.5.1.3	4.9.15
Tuning control	3.5.3	4.9.16
Electromagnetic interference	3.8	4.9.17
Subgroup 2		
Temperature	3.7.1	4.9.1
Altitude	3.7.3	4.9.3
Rain	3.7.5	4.9.10
Vibration (ground vehicle)	3.7.4	4.9.4
Shock (basic design)	3.7.4	4.9.5
Subgroup 3		
Humidity	3.7.2	4.9.2
Immersion	3.7.6	4.9.11
Fungus (NOTE 1)	3.7.9	4.9.8
Loose cargo bounce	3.7.4	
Subgroup 4		
Dust (NOTE 1)	3.7.7	4.9.12
Salt fog (NOTE 1)	3.7.8	4.9.13
Vibration (secured cargo)	3.7.4	4.9.6
Bench handling	3.7.4	4.9.7
Transit drop	3.7.4	4.9.9

NOTE 1: The equipment shall be thoroughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent tests.

TABLE VI - Group D inspection

Inspection	Req. Para.	Test Para.
Reliability	3.9	4.11

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4.8 Electrical tests.

4.8.1 Standard test conditions. The electrical tests shall be conducted under the following conditions:

a. Unless otherwise specified, all tests shall be conducted under conditions of normal room ambient temperature and humidity.

b. Unless otherwise specified, all tests shall be conducted using the nominal 115V input, however the various rated input voltages are as follows:

(1)	Vehicular supply	26 ± 0.5 Volt dc
(2)	115 Volt AC supply	115 ± 2V, 60 Hz
(3)	230 Volt AC supply	230 ± 4V, 60 Hz
(4)	Dry battery supply	18 Volts ± 0.5 Volt dc

c. RF input - signal generator with 50 ohm output impedance using a 50 ohm pad in series.

d. Modulation

AM - - - - - 30% at 1000 Hz

e. Audio output load impedance - - - - 600 ohms

f. Audio output level 10 mW, unless otherwise specified.

4.8.2 Detailed electrical tests. The equipment shall meet the applicable requirements of section 3 of this specification when tested in accordance with the procedures of SM-A-747539 as identified in TABLE VI, below:

4.8.2.1 Input impedance. In addition to the test described by para. 6.6 of SM-A-747539, measure input impedance with a spectrum analyzer, sweeping the entire frequency range.

4.8.2.2 Battery supply. In addition to the tests described by SM-A-747539, read the voltage available at dust cover J1, pin 4 when 12 fresh BA-30/U or BA-3030/U cells are in place. It shall be ≥18Vdc.

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TABLE VI - Performance tests.

Test Title	Req. Para.	SM-A-747539 Para. No.	Test Group
Input impedance	3.6.1.1	6.6 of SM-A-7747539 and 4.8.2.1 of this spec.	B
Antenna input isolation	3.6.1.2	6.6	B
Frequency range (overlap)	3.6.2	5.6	A
Freq. stability vs. time	3.6.3.1	6.7	B
Freq. stability vs. internal battery voltage	3.6.3.2	6.8	B
Freq. stability vs. received signal level	3.6.3.3	6.9	B
Freq. stability vs. temperature	3.6.3.4	7.8.1	C
Freq. stability vs. humidity	3.6.3.5	7.9.2	C
AM sensitivity	3.6.4.1	5.7	A
CW sensitivity	3.6.4.1	5.8	A
FM sensitivity	3.6.4.1	5.9	A
AM whip antenna sensitivity	3.6.4.2	6.10	B
AM sensitivity vs. internal battery voltage	3.6.4.3	6.11	B
AM & CW sensitivity vs. temperature	3.6.4.4	7.8.3	C
FM sensitivity vs. temperature	3.6.4.5	7.8.4	C
AM & CW sensitivity vs. humidity	3.6.4.6	7.9.3	C
FM sensitivity vs. humidity	3.6.4.7	7.9.4	C
Freq. calibrator accuracy	3.6.5.1	5.10	A
Freq. calibrator degradation	3.6.5.2	7.8.5	C
Injection level	3.6.5.3	5.11	A
Spurious indications	3.6.5.4	5.12	A
Calibration adjustment	3.6.5.5	6.5	B
Dial readout error	3.6.6	5.13	A
Tuning backlash	3.6.7	5.4	A
Image rejection	3.6.8.1	6.12	B
IF rejection	3.6.8.2	6.13	B
Crossmodulation	3.6.8.3	6.14	B
Intermodulation distortion prod.	3.6.8.4	6.15	B
IF center frequency	3.6.9.1	6.16	B
IF selectivity	3.6.9.2	6.17	B
AGC characteristic	3.6.10	5.15	A
Limiter characteristic	3.6.11	5.16	A
Beat frequency oscillator	3.6.12	5.17	A
Audio power output	3.6.13.1	5.18	A
Audio harmonic distortion	3.6.13.2	5.19	A
Audio response	3.6.13.3	6.18	B

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TABLE VI - Performance tests (Cont'd)

Test Title	Req. Para.	SM-A-747539 Para. No.	Test Group
Microphonics	3.6.13.4	6.19	B
Diode Output level	3.6.14.1	5.20	A
Discriminator Output	3.6.15	5.14	A
Input power requirements	3.6.16.1	5.21	A
Voltage regulation	3.6.16.2.2	5.22	A
Output voltage ripple	3.6.16.2.2	5.23	A
Battery supply	3.6.16.3	4.8.2.2 of spec.	B
Solid state device protection	3.6.16.4	7.7.1	C
Reverse polarity protection	3.6.16.5	7.7.2	C
Panel meter	3.6.16.6	5.3	A
Dial light	3.6.17	5.5	A
Final operation	3.6.18	5.24	A

4.9 Environmental tests. All environmental test methods, conditions and procedures given below are contained in MIL-STD-810C. The equipment performance measurements shall be made under standard conditions as defined in MIL-STD-810C. Successful completion of any particular test shall be considered evidence of meeting the corresponding service condition.

4.9.1 Temperature. The equipment shall be tested according to the following test methods.

4.9.1.1 Low temperature. The equipment shall be tested in accordance with Method 502.1, Procedure 1 of MIL-STD-810C and shall meet the requirements of 3.7.1.1.

Storage temperature (Step 2)	-65°F (-54°C)
Operating temperature (Steps 4, 5)	-40°F (-40°C)
Make measurements at Steps 1, 5, 7	

4.9.1.2 High temperature. The equipment shall be tested in accordance with Method 501.1, Procedure II of MIL-STD-810C and shall meet the requirements of 3.7.1.2.

Storage temperature (Step 4)	+160°F (+71°C)
Operating temperature (Steps 7, 8)	+154°F (+68°C)
Make measurements at Steps 1, 8, 10 (Temperature at steps 2,3 & 5)	+120°F (+68°C)

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4.9.2 Humidity. The equipment shall be tested in accordance with Method 507.1, Procedure III of MIL-STD-810C. In Step 7, the chassis shall be removed from the dust cover so as to expose the normally sealed areas to the chamber environment. The equipment shall be operated during the last 5 hours of the last cycle of Step 6 and shall meet the requirements of 3.7.2 when tested in Steps 4, 6 (last cycle only), 8 and 11.

4.9.3 Altitude. The equipment shall be tested in accordance with Method 500.1, Procedure I of MIL-STD-810C. A non-operating test shall be performed for 1 hour at 40,000 feet. The equipment shall meet the requirements of 3.7.3 when tested at Steps 1, 2 and 4.

Operating altitude	15,000 feet
Non-operating altitude	40,000 feet

4.9.4 Vibration (ground vehicle installation). The receiver, when mounted on its Shock Mount Base, MT-4034/UR, shall be tested in accordance with Method 514.2, Procedure VIII of MIL-STD-810C and shall meet the requirements of 3.7.4.

Equipment category	F
Applicable table	514.2 - VI
Applicable figure	514.2 - 6
Test curve	V (1.5g)
Sweep	12 min., 5-200-5Hz
Cycle time	3 hrs. each, 3 directions

4.9.5 Shock (basic design). The receiver, when mounted on its Shock Mount Base, MT-4034/UR, shall be tested in accordance with Method 516.2, Procedure I of MIL-STD-810C and shall meet the requirements of 3.7.4.

Equipment category	ground vehicle
Applicable figure	516.2-1
Peak shock value	40 g
Duration	11 ms

4.9.6 Vibration (secured cargo transportation). The receiver alone, without Shock Mount, shall be tested in accordance with Method 514.2, Procedure X of MIL-STD-810C and shall meet the requirements of 3.7.4.

Transport mode	rail, air, sea, truck or semi
Applicable table	514.2 - VII
Applicable figure	514.2-7
Test curve	AW (1.5g)
Sweep	12 min., 5-200-5Hz
Cycle time	84 min. each, 3 axes

4.9.7 Shock (bench and handling). The equipment shall be tested in accordance with Method 516.2, Procedure V of MIL-STD-810C, and shall meet the requirements of 3.7.4.

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4.9.8 Shock (loose cargo bounce). The equipment shall be tested in accordance with Method 514.2, Procedure XI, Part 2 of MIL-STD-810C, and shall meet the requirements of 3.7.4.

4.9.9 Shock (transit drop). The receiver, in the canvas bag, shall be tested in accordance with Method 516.2, Procedure II, of MIL-STD-810C, except that the drop height shall be 30 inches. Receiver only shall be in canvas bag during test. Prior to and after testing the equipment shall meet the requirements of 3.7.4.

4.9.10 Rain. The receiver, excluding the canvas bag, shall be tested in accordance with Method 506.1, Procedure I of MIL-STD-810C, and shall meet the requirements of 3.7.5.

4.9.11 Immersion. The receiver, excluding the canvas bag, shall be tested in accordance with Method 512.1, Procedure I of MIL-STD-810C, and shall meet the requirements of 3.7.6.

4.9.12 Dust. The receiver, excluding the canvas bag, shall be tested in accordance with Method 510.1, Procedure I of MIL-STD-810C, and shall meet the requirements of 3.7.7.

4.9.13 Salt fog. The equipment, including the receiver and all accessories (the receiver not in canvas bag during salt atmosphere exposure), shall be tested in accordance with Method 509.1, Procedure I of MIL-STD-810C, and shall meet the requirements of 3.7.8.

4.9.14 Fungus. The equipment in the assembled and as ready for delivery condition, or representative samples subject to approval of the procuring agency, shall be subjected to the test of MIL-STD-810C, Method 508.1, Procedure I, except that the temperature humidity conditions of paragraph 3.1.6a shall be changed to a constant 29°C (84°F) and 95% RH (minimum) during the life of the test. The equipment or samples shall not be specially cleaned for the fungus test, except for the cleaning it receives during or after production. All enclosed or gasketed assemblies shall be opened and the interior sprayed with the specified mixed spore suspension. After 28 days, based on visual examination, the equipment assembly or samples shall show no more than sparse microbial growth with restricted tuberal growth development in an area 10% or less of the total area and no more than six unrelated minute colonies with mycelial development in areas only in other than critical circuit portions, such as terminal spacing, printed circuit boards, etc., with sparse growth due to random contamination or traces of unmixed material ingredients. The equipment shall fail if it does not meet criteria of 3.7.9 or if it shows more than the growth specified above.

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4.9.15 Canvas shrinkage. The canvas bag shall meet the requirements of 3.5.1.3 before and after testing in accordance with SM-A-747539 paragraph 7.3.

4.9.16 Tuning control. The tuning control of the receiver shall meet the requirements of 3.5.3 when tested in accordance with SM-A-747539 paragraph 7.5.

4.9.17 Electromagnetic interference. The equipment shall be tested in accordance with paragraph 7.16 of SM-A-747539 and the test setups and measurement techniques of MIL-STD-462, Notice 3 as implemented by a contractor prepared Government approved EMI test plan and shall meet the requirements of 3.8. Exclude tuned channel frequencies for power line conducted susceptibility measurements of CS02.

4.10 Interchangeability. The equipment shall be inspected to insure the physical and functional interchangeability requirements of 3.5.8.

4.10.1 Electrical interchangeability. All like modules and replaceable parts shall be tested for functional interchangeability by assembly of randomly selected parts. Following the random assembly process, the equipment shall be tested to and shall meet Group A acceptance requirements. Electrical retuning or realignment of modules during production assembly may be expected and are permissible. Random selection for this assembly shall be supervised by the Government.

4.10.2 Mechanical interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirements of 3.5.8. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of case, cover and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as case, covers and mountings.

d. Location of connectors, locking pins, fasteners, slides and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

f. Compatibility of switching on RF circuit boards and shaft.

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4.11 Reliability. The equipment shall be tested in accordance with test plan VC of MIL-STD-781C to demonstrate a lower test MTBF of 720 hours and shall meet the requirements of 3.9. The environmental conditions and duty cycle shall be in accordance with SC-A-678020.

4.12 Quality conformance inspection for packaging. Quality conformance inspection for packaging shall be in accordance with Group A, Group B and, when required (see 6.2.c), the Group C (rough handling) requirements of MIL-P-116. Inspection lots shall be in accordance with MIL-P-116.

4.13 Rough handling test (packaging). When the rough handling test of specification MIL-P-116 is required by the contract (see 6.2.c), the AM sensitivity and diode output requirements of 3.6.4.1 and 3.6.14 shall be met to determine freedom from operational malfunction caused by rough handling.

5. PACKAGING.

5.1 Packaging. Packaging shall be in accordance with SPI 1G00075.

6. NOTES.

6.1 Intended use. The AN/URR-71 receiving set is a VHF superheterodyne set capable of receiving AM, FM, CW and MCW signals in two bands over a frequency range of 19.0 to 157.5 MHz. While it is capable of fixed station, mobile or manpack use, its primary intended uses include direction finding in connection with the AN/TRQ-30, AN/TRQ-32 and AN/TRR-33. The flat top, relatively vertical side IF skirt makes this receiver ideal for DF work.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level A or Level B preservation and packing (see section 5).

c. When rough handling test is required (see 4.13).

d. First article inspection: Four sample units of each item cited in Section 1 are generally required to reduce the overall time for lengthy environmental and performance tests (may be modified by contract).

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6.2 Ordering data (Cont'd).

e. Marking and shipping of samples.

f. Data requirements. Deliverable data required by this specification is cited in 4.4.3 entitled First Article Data.

g. Place of final inspection.

h. Technical literature required.

i. Quantity of tools and running spare parts required.

6.3 Location of operational inspection. It is desirable that the operational inspection (3.6.18/4.8.2) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.4 Color. The color chip furnished by the procuring agency (see 3.5.1.1), will be color chip No. X-24087, and may be obtained upon request to Commanding Officer, U.S. Naval Depot, 5801 Tabor Avenue, Philadelphia, PA.

6.5 Color textiles (see 3.5.1.3.1). Standard samples of Olive Drab No. 7 may be obtained from the Sample Loan Unit, Military Clothing and Textile Supply Agency, Philadelphia, Pennsylvania. When requesting samples, indication should be given as to type of material concerned.

6.6 Definitions.

6.6.1 Temperature stabilization. Temperature stabilization will have been attained when the temperature of the part of the test item considered to have the longest thermal lag does not change more than 2.0°C (3.6°F) per hour.

6.6.2 Physical damage. Physical damage is defined as corrosion, rust, blistering, swelling or other deterioration of parts and materials.

6.7 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

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6.8 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-R-55319A(CR) Receiving Set, Radio AN/URR-71

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